Appendix README

Capstone Client Handoff - Appendix Research

Purpose

This appendix contains background notebooks that document dataset creation, parser support, early exploratory analysis, and preliminary modeling work. These files are included for transparency and continuity, but final reported results should be taken from the final notebooks and demo materials in the main handoff folders.

Primary Source of Final Results

For final client-facing conclusions, use the locked notebooks in 03_Final_Notebooks and the demo materials in 05_Client_Demo_Materials. Appendix notebooks are intended to explain how the project developed and how supporting work was performed.

Use Note

Some intermediate work from earlier project stages was intentionally left out because it used superseded or unreliable data versions. If a number in an appendix notebook differs from a final notebook, the final notebook value should be used for reporting and client interpretation.

Appendix Files

Appendix File Name	Purpose
Dataset_Creation_and_Validation_Workflow.ipynb	Step-by-step workflow showing how the raw DSSAT .OUT files were parsed, cleaned, merged, validated, and saved during the final dataset creation stage.
DSSATout2Dataframe.ipynb	Parser support notebook used to convert DSSAT output files into tabular data before cleaning and merging.
Early_Project_EDA_and_Modeling.ipynb	Early project exploration, initial modeling, and broad analysis work.
TaskB_LightGBM_Preliminary_Dataset_Exploration.ipynb	Early LightGBM exploration using a preliminary dataset before the final validated dataset was available.

How to Use These Files

- Use the final notebooks folder for final reported model results.

- Use the client demo folder for presentation-ready walkthroughs and HTML views.

- Use Dataset_Creation_and_Validation_Workflow.ipynb if the client wants to verify or recreate the final validated dataset from raw DSSAT outputs.

- Use this appendix only when you want to understand how the project developed over time or review supporting exploratory work.

Folder Placement

Place this document in: 06_Appendix_Research/